Week-5
Numerical Problems

Q.2.

$$m = 0.9 \quad \left(m = \frac{2.7}{3} \right)$$

$$L = \sqrt{\frac{\left(\frac{m V_o}{2}\right)^2 - V_{s,pk}^2}{\left(2\pi f_s I_{s,pk}\right)^2}}$$

$V_o = 750\,V$; $V_{s,pk} = \frac{400}{\sqrt{3}} \times \sqrt{2}$; $f_s = 50\,Hz$,

$$P = \sqrt{3} \cdot V_{L-L,Rms} \cdot I_{Rms}$$

$$\Rightarrow I_{s,rms} = \frac{22000}{\sqrt{3} \times 400} = 31.75$$

$$\Rightarrow I_{s,pk} = 31.75 \times \sqrt{2}$$
$$= 45A$$

$$\Rightarrow L = \frac{\sqrt{\left(\frac{0.9 \times 750}{2}\right)^2 - (326.59)^2}}{\sqrt{\left(2 \times 3.14 \times 50 \times 45\right)^2}}$$

$$L = \frac{85.08}{14130} = 6.02\,mH$$

Q.5.

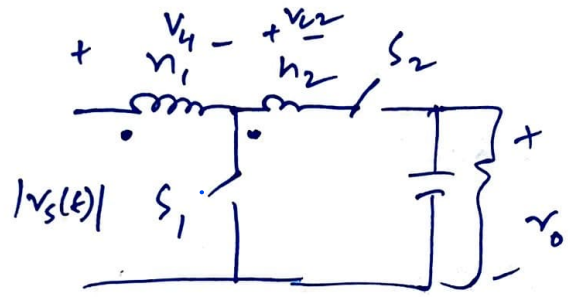
given carrier frequency = 10 kHz

fundamental freq. = 50 Hz

⇒ Possible answer would be ⇒ $1.10000 + 2.50 = 10100$

Q.6.

Given circuit



⇒ $V_L = V_{L1} + V_{L2}$

$d(t)\,T_s$

S_1 is on ; S_2 is off

⇒ $V_{L1} = |V_s(t)|$; $V_{L2} = \frac{n_2}{n_1} \cdot |V_s(t)|$

$V_L = V_{L1} + V_{L2}$

⇒ $V_L = |V_s(t)| + \frac{n_2}{n_1} |V_s(t)|$

$(1 - d(t))\,T_s$

S_2 is on , S_1 is off

⇒ $V_L = |V_s(t)| - V_o$

Apply volt-sec Balance

$$\frac{1}{T/2}\int_{0}^{T/2}\left[\left(|v_s(t)| + \frac{n_2}{n_1}|v_s(t)|\right)d(t)\cdot T_s + \left(|v_s(t)| - v_o\right)\left(1-d(t)\right)T_s\right]\cdot dt = 0$$

Apply approximations,

$$\left(|v_s(t)| + \frac{n_2}{n_1}|v_s(t)|\right)d(t) + \left(|v_s(t)| - v_o\right)\left(1-d(t)\right) = 0$$

$$\Rightarrow \cancel{|v_s(t)|\,d(t)} + \frac{n_2}{n_1}|v_s(t)|\cdot d(t) + |v_s(t)| - v_o - \cancel{d(t)|v_s(t)|} + v_o d(t) = 0$$

$$\Rightarrow d(t)\left(v_o + \frac{n_2}{n_1}|v_s(t)|\right) = v_o - |v_s(t)|$$

$$\Rightarrow d(t) = \frac{v_o - |v_s(t)|}{v_o + \frac{n_2}{n_1}|v_s(t)|}$$

Answer: B.

Q.7.

$$d(t) = \frac{1}{1 + \frac{|v_{s,pk}\sin\omega t|}{nv_o}} \quad \text{(from Lec-21)}$$

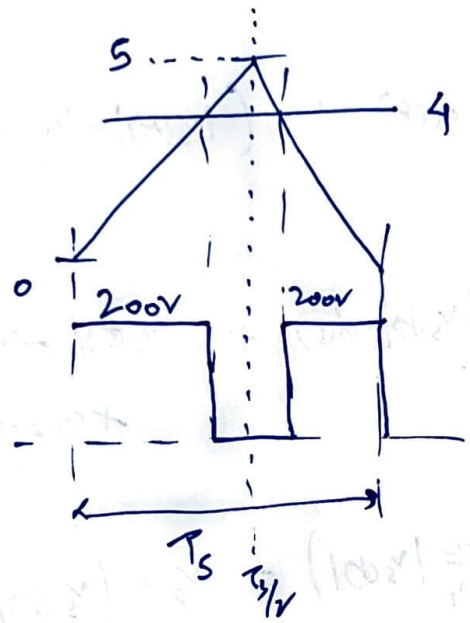
$$0.12 = \frac{1}{1 + \frac{230\sqrt{2}}{0.6\times v_o}} \Rightarrow \frac{230\sqrt{2}}{0.6\times v_o} = \frac{1}{0.12} - 1$$

$$\cancel{v_o = 230\sqrt{2}\times 0.12}$$

$$\Rightarrow V_o = \frac{230 \times \sqrt{2}}{0.6 \times 7.33}$$

$$\boxed{V_o = 73.958}$$

Q. 8.



In T_s duration, apply ~~voltage~~ averaging

$$V_{o,avg} = \frac{200 \times 0.8 \times \frac{T_s}{2}}{T_s/2}$$

$$V_{o,avg} = 160\ V$$

Q.9.

$$d(t) = \frac{1}{1 + \frac{M_{sp} \sin \omega t}{n V_o}}$$

$$d(t) = \frac{1}{1 + \frac{2 \times 0.8 \times \sqrt{2}}{2 \times 400}}$$

$$d(t) = 0.702$$